

From Green Intentions to Green Implementations

Solar, wind, mechanical, hydroelectric, geothermal... there are lots of green energy alternatives available today that can provide energy for a small watch to a town's power grid. From sub-\$100 do-it-yourself kits to mega-budget deep-water turbines, there are a million means to implement these



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Seems like a clichéd way to start such an article, but still let me repeat an oft-quoted Native American proverb: “We do not inherit the earth from our ancestors. We borrow it from our children.” Unfortunately, we realised this only after inventing every type of power-guzzling equipment possible—right from power plants that swallow lignite from the earth’s belly, to PCs and mobiles that scrape little morsels of energy from the bowl, albeit summing up to a mega-sized carbon footprint. Temperatures have risen, glaciers have melted, fertile lands have become parched, wind directions have changed, everything short of the sun rising in the

west has happened, thanks to the way we have been exploiting the earth’s resources to feed our perennially hungry devices! Yet, the truth is we cannot live without these devices.

Fortunately, realisation about the mishap caused by our new-fangled technologies has been accompanied by the development of interesting ‘green’ techniques that can meet our power needs in more eco-friendly ways, characterised by greater sustainability, and lower pollution and carbon emissions. Solar, hydro, geothermal, wind, biomass—there are a lot of options available today that can, depending on the technology and implementation, provide energy at various scales to meet requirements of electronic devices to electricity grids. These sources

have been around for long, but the processes and technologies for tapping these sources have become more effective and efficient only recently.

Energy saving too has become very important. “We are becoming more and more aware of how important it is for us to be energy-conscience, and how if directly affects our environment. Efficient design of data centres to reduce power consumption, energy-efficient glass in buildings to reduce the load on ACs, LEDs for energy-efficient lighting and displays, power management chips, low-power libraries to manufacture semiconductor chips, software for low-power applications, and efficient inverters and other electronics for energy generation and distribution are some important areas,” says Dr Madhusudan Atre, president and managing director, Applied Materials India.

While technology is not the only aspect of green energy, a complete green energy initiative also involves efficient usage of the energy, green certificates, other national initiatives to promote the concept and so on—it certainly is a focus area. In fact, there are many levels at which technology plays an important role in the green energy space—technologies that help tap green energy sources in an efficient way, technologies that help augment this energy with a larger grid, technologies that help channel this energy to power smaller devices and so forth.

There have been significant developments in every such dimension of green